19 · Reading the Paper Critically

A good way to understand a paper is to ask what it *didn't* show. This file collects the paper's own stated limitations and a few extra points that are worth checking yourself.

Step 1: Limitations the paper states

Limitation	Why it matters
One dataset (QReCC)	Other domains, languages or document types may behave differently.
Same-family models (Qwen3-8B → Qwen3-Embedding-8B)	Siblings may share internal structure. Mapping into an unrelated embedder is probably harder.
Still worse than baseline, significantly	It's a trade: speed and simplicity for a few points of quality.
Teacher needed for training	Only deployment is simplified.
Unknown generalization to new documents	Not tested whether the head holds up when the collection changes.

Step 2: The motivation vs. the evidence

The strongest argument in the introduction is Problem 3: hidden states are *richer* than query text. But:

- The student is trained to **match the teacher**, and the teacher only sees query text. So training rewards reproducing the text-based vector, not going beyond it.
- No experiment tests whether the hidden states actually carry extra, useful context (for example, a variant trained on

relevance labels, or one that also reads hidden states from the conversation itself).
- The result, slightly *below* the teacher, is what you would expect from good imitation, not from exploiting extra information.

So the paper shows **"nearly as good, much cheaper"**. It doesn't yet show **"better because of richer context"**. That is an open and interesting next step.

Step 3: Numbers worth double-checking

These are not necessarily errors; they are places where the text is ambiguous, and a careful reader should notice them.

Item	What the paper says	What to notice
"Retains 97%"	Headline claim; "95–97%" in Section 5.1.	The gaps (−3.0, −3.6, −3.5) are in points. Relative retention is 95.3% (Recall), 89.1% (MRR), 91.3% (nDCG).
Test set size	2,189 retrieval triggers in the whole split, divided into train/val/test by conversation.	The test win/tie/loss counts add up to 140 + 1843 + 206 = **2,189**, the full number, and the per-conversation analysis covers all **346** conversations. Either the test set is the full set, or the counts are over everything. It isn't clear which.

Item	What the paper says	What to notice
Latency	43.5 ms vs 2.0 ms, 21.8×.	Compares only the encode step. Both pipelines still generate the query with the 8B LLM, which usually takes much longer than 43.5 ms. End-to-end speedup is far smaller than 21.8×.
"Avoid text generation for retrieval altogether"	Said when contrasting with HyDE.	The method still generates the query text; its hidden states come from exactly that generation. It avoids *extra* generation (compared to HyDE) and *re-encoding*, not generation.
~25M parameters	Section 4.4.	Counting the listed layers (4096→1024 input, 2 layers with FFN 4096, 1024→1024 output, 128 positions) gives about 30.6M. Close enough to be "tiny", but not 25M.
Randomness	Bootstrap CIs over test triggers.	These capture test-set noise, not training noise. There is no mention of multiple training seeds, which matters because training is sensitive (lr 5e-4 collapses).

Step 4: Questions to think about

1. Would a cross-family pair (say, a Llama LLM and a different embedder) still reach 95%?
2. Could training on real relevance labels, in addition to the teacher, let the student *beat* the teacher?
3. Which hidden layer is best? Only the last one was tried.
4. Would attention pooling or last-token pooling beat mean pooling?
5. Could the query text be skipped entirely by projecting from the hidden state at the moment the model *decides* to search?

Step 5: The fair summary

The paper identifies a real redundancy, proposes a simple and cheap fix, and shows with careful ablations and proper statistics that a ~25–30M head can stand in for an 8B embedder at a small quality cost, with a large saving in the encode step. The bigger promise, that searching with the LLM's own thoughts could beat searching with its words, is still to be demonstrated.